USER GUIDE

From a blank workspace to a checked class.

Tsek prints answer sheets, reads the shaded bubbles, adds the scores you type for handwritten parts, and turns the whole class into one clean record. This guide walks the same 12-learner class through every tab, one step at a time.

19 screenshots · Windows desktop + browser · Works offline · No accounts, no uploads

First built for San Roque Parish High School · open project

ZX TechLabs · Richmond C. Jabla

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Before you start

Tsek is a paper workflow. The better the paper, the fewer sheets you have to correct by hand — so set these up once, properly.

WHAT YOU NEED

- ☐ A printer that can print at **100% / actual size**.
- ☐ Plain white paper, in the size you choose in Setup.
- ☐ A scanner, or a phone camera that makes PDF, JPG, or PNG files.
- ☐ Tsek installed on Windows, or open in a browser.

THREE HABITS THAT SAVE TIME

- ☐ Finish the structure and key before printing anything.
- ☐ Print at 100% — "Fit to page" shrinks the corner marks.
- ☐ Back up the workspace before wiping app data or moving machines.

THIS GUIDE USES THE DEMO CLASS

On a fresh install, choose `Explore the demo` on the home screen. It fills in a 12-learner class and both sample exams, so the screenshots and numbers here match what you see. You can delete the demo later.

How this guide works

Each numbered chapter follows one tab, in the order you would normally use them. The circled number on the left matches the step, not the tab count.

`Setup` → `Roster` → `Answer key` → `TOS` → `Sheets` → `Scan` → `Scores` → `Results` → `Analysis` → `Export`

- **Figures 1–19** are real screenshots. On screen, click one to open it full size.
- **Rust boxes** are notes about how Tsek behaves. **Green boxes** are shortcuts and tips. **Red boxes** are warnings that can cost you a scan or a score.
- Numbers in the text come from the demo class, so you can compare them with your own screen at each step.

1 Open the workspace

The home screen answers one question before you open anything: how far along is each class?

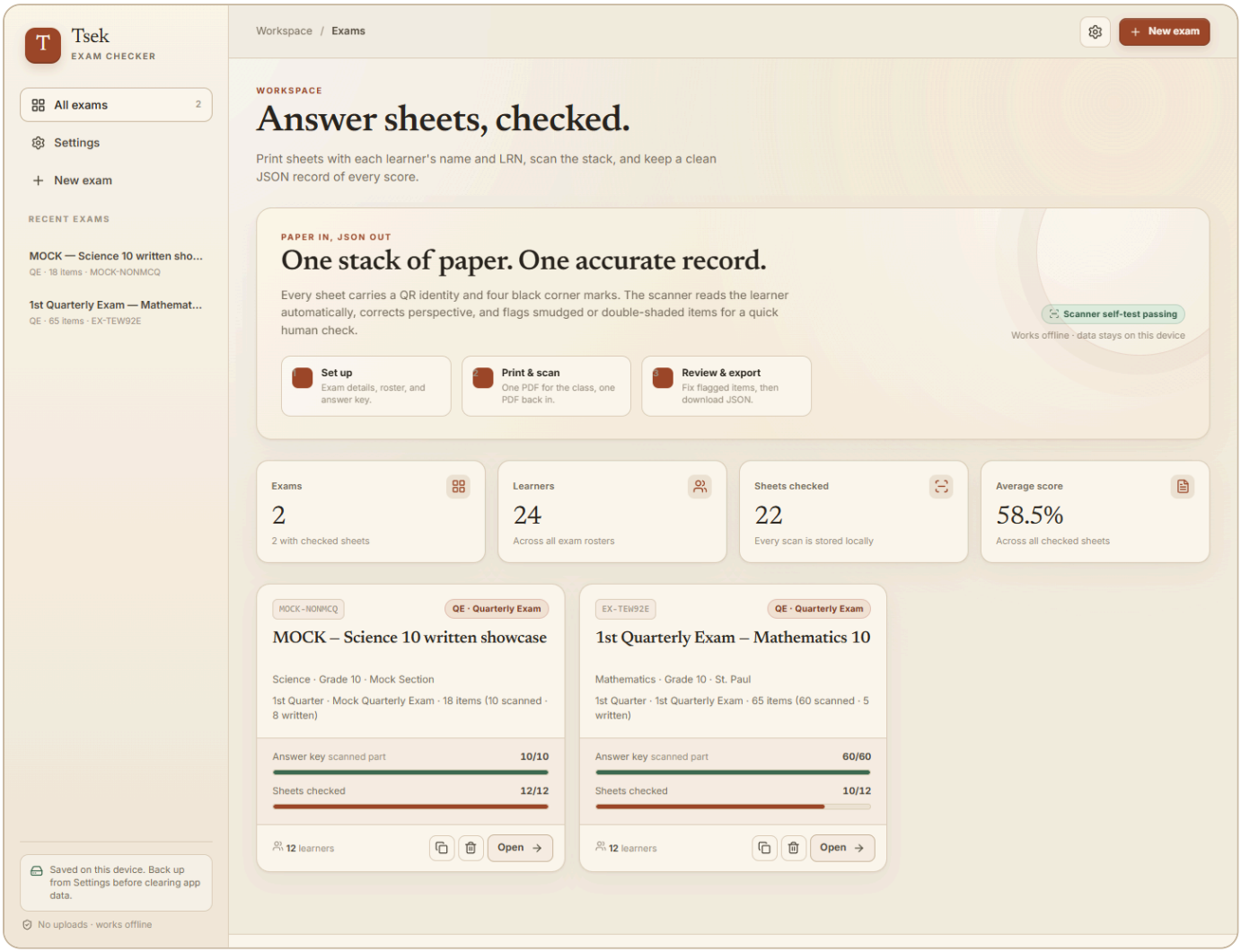


FIGURE 1 All exams is the home screen. Two exam cards show answer-key progress and how many sheets have been checked.

1. Open Tsek. **All exams** is selected in the left rail.
2. Read the four tiles: **Exams**, **Learners**, **Sheets checked**, and **Average score**. They cover every exam in the workspace, not just one.
3. Check each exam card. **Answer key scanned part** tells you whether a key is complete; **Sheets checked** tells you how much of the class has been scanned.
4. Use `Open` on a card to enter the exam, or `New exam` to start a fresh class.

In the demo, the MOCK exam has a complete key and 12 of 12 sheets checked. The Mathematics exam is still at 10 of 12, which is the more realistic state on a normal school day.

> **PROGRESS, NOT DECORATION**
>
> If a card says 0 of 12 sheets, the problem is usually printing or scanning — not scoring. Fix that first; analysis will stay blank until papers exist.

2 Create the exam and fill in the details

The words on this screen are printed on every answer sheet and carried into every export. It is worth three minutes to get them right.

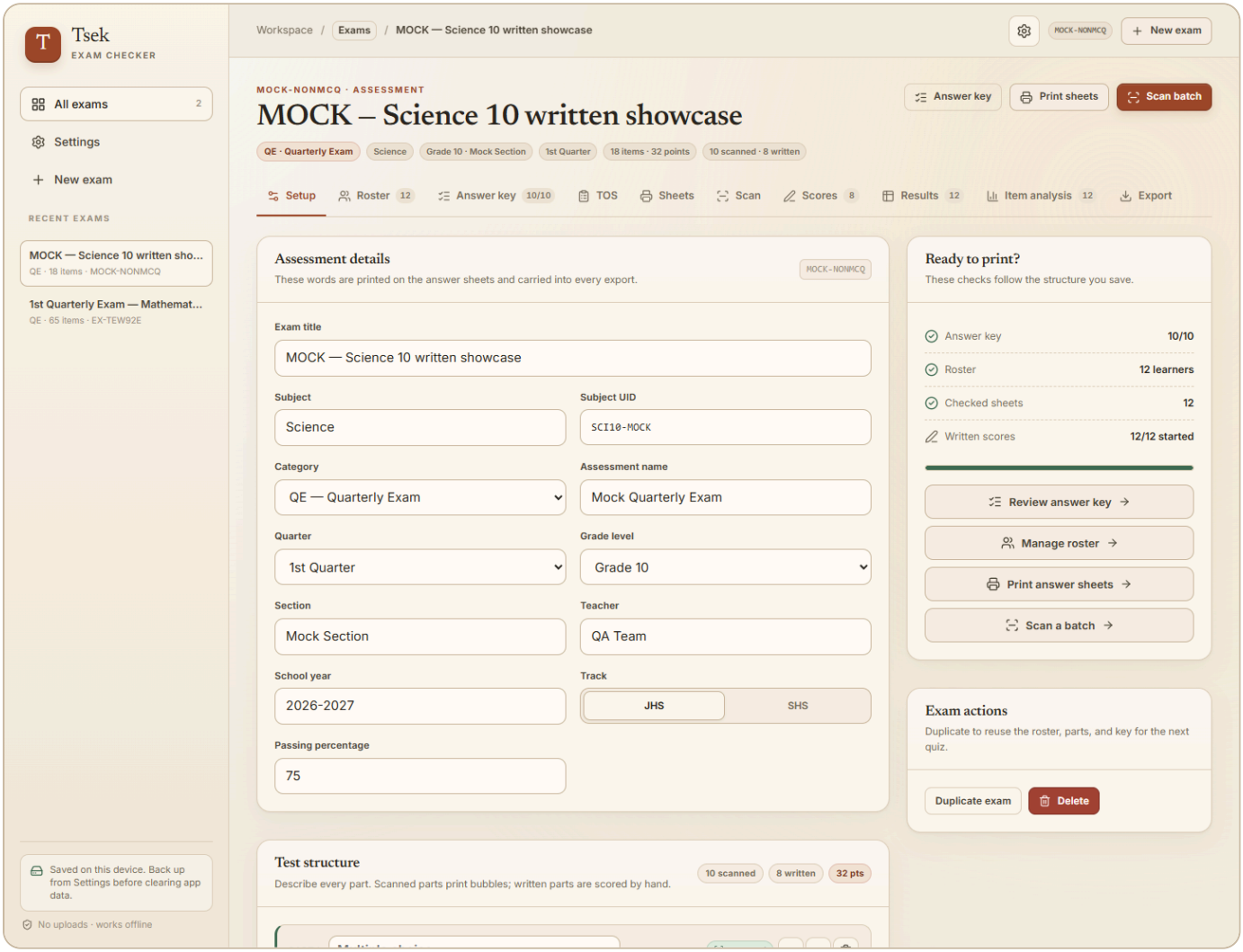


FIGURE 2 The Setup tab for the MOCK exam. The checklist on the right is computed from the structure, roster, scans, and written scores.

1. Choose `New exam`, then pick a category: **Written Works**, **Performance Task**, or **Quarterly Exam**. The category decides which weight the GradeApp applies later, so it matters even if the test itself looks the same.

2. Fill **Exam title**, **Subject**, and **Subject UID**. The UID is the short code the GradeApp uses to match the subject; keep it consistent across quarters.

3. Set **Category**, **Assessment name**, **Quarter**, **Grade level**, **Section**, **Teacher**, **School year**, and **Track** (JHS or SHS).
4. Set **Passing percentage**. The default is 75; this drives the passing count and the grade check in Results.
5. Watch the **Ready to print?** panel. Each line turns into a check as that piece is finished: key, roster, checked sheets, written scores.
6. Save changes before leaving the tab. Text edits here do not change the scanned geometry, so correcting a typo does not invalidate papers.

> **DUPLICATE INSTEAD OF REBUILDING**
>
> For the next quarter, use Duplicate exam under Exam actions. It copies the roster, parts, and key so you only change what is new. Deleting asks you to type a confirmation and saves a snapshot first.

3 Describe the test structure

A test is a list of named parts. Each part is either scanned from bubbles or scored by hand. This one screen is the source of the printed sheet, the TOS, the score columns, and the analysis.

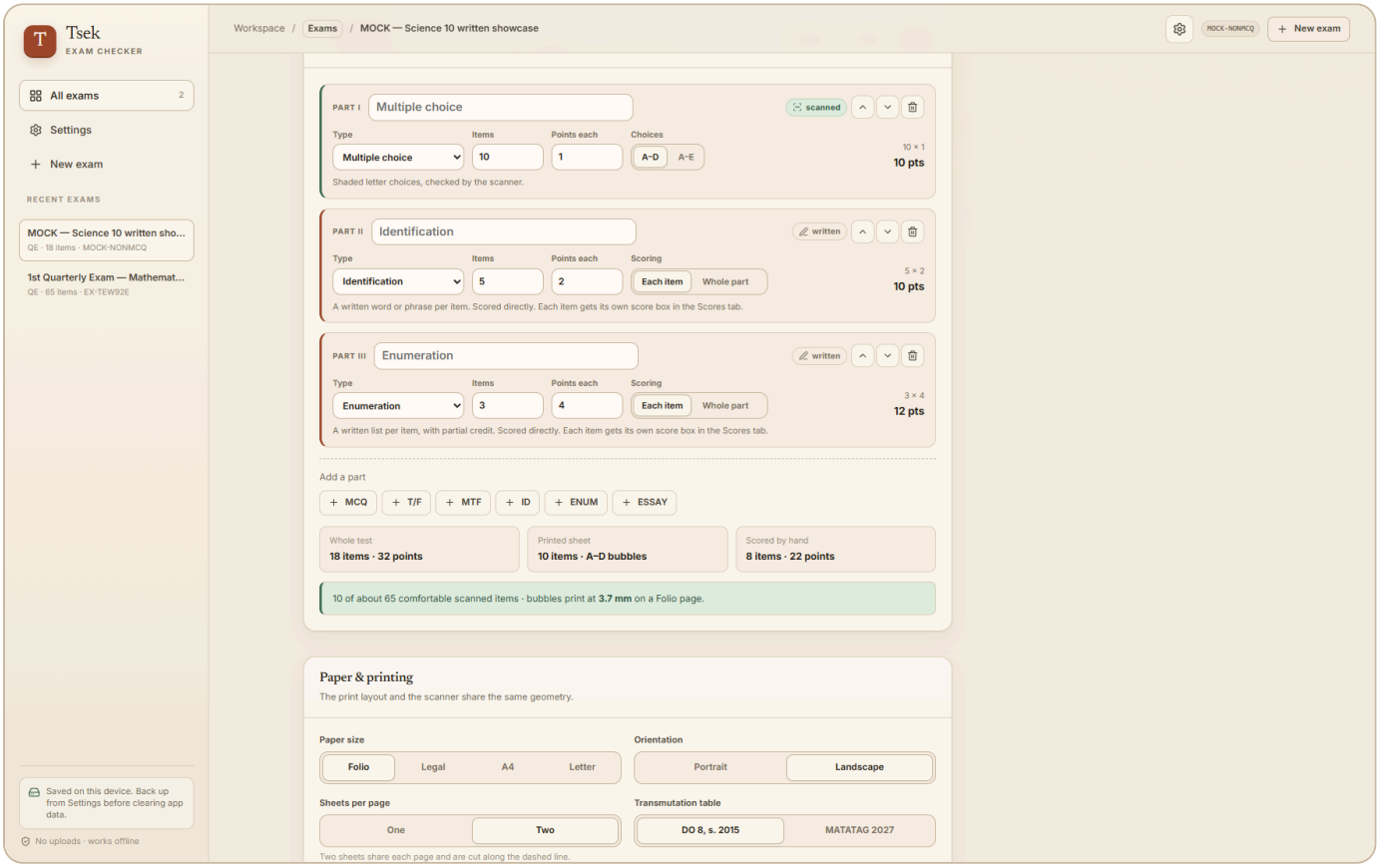


FIGURE 3 Three parts: 10 scanned multiple-choice items, 5 identification items, and 3 enumeration items. Totals confirm 18 items and 32 points.

1. Under **Test structure**, name each part and choose its **Type**. Multiple choice, true/false, and modified true/false are marked **scanned**; identification, enumeration, and essay are marked **written**.
2. Set **Items** and **Points each**. The part total updates immediately; the whole-test total is the sum of every part.
3. For a written part, choose how it is scored: **Each item** gives one score box per item; **Whole part** gives one box for the entire answer. See step 12 for when to use which.
4. Use the add-a-part buttons (**MCQ, T/F, MTF, ID, ENUM, ESSAY**) to extend the test. Reorder or delete a part with the arrows and bin on its header.
5. Read the three summary boxes: **Whole test**, **Printed sheet**, and **Scored by hand**. They are an easy sanity check before printing.
6. Set **Paper & printing**: paper size, orientation, sheets per page, and the transmutation table. The print layout and the scanner share the same geometry.

BUBBLE DENSITY

Tsek warns when a page is getting crowded. In the demo, 10 scanned items print 3.7 mm bubbles; the Mathematics example prints 3.5 mm. About 65 scanned items is the comfortable ceiling on a Folio page at two sheets per page. Split very long tests into parts on separate pages rather than shrinking the bubbles.

CHANGING STRUCTURE AFTER SCANNING

Changing the **item count** clears scanned results for that exam, because old bubbles no longer line up with new items. Changing paper, orientation, or sheets per page keeps the scores but changes the printed geometry — reprint before scanning again.

4 Import the class roster

The LRN is the learner's identity. Names are printed for the teacher and learner; the QR carries the LRN, and that is what the scanner matches.

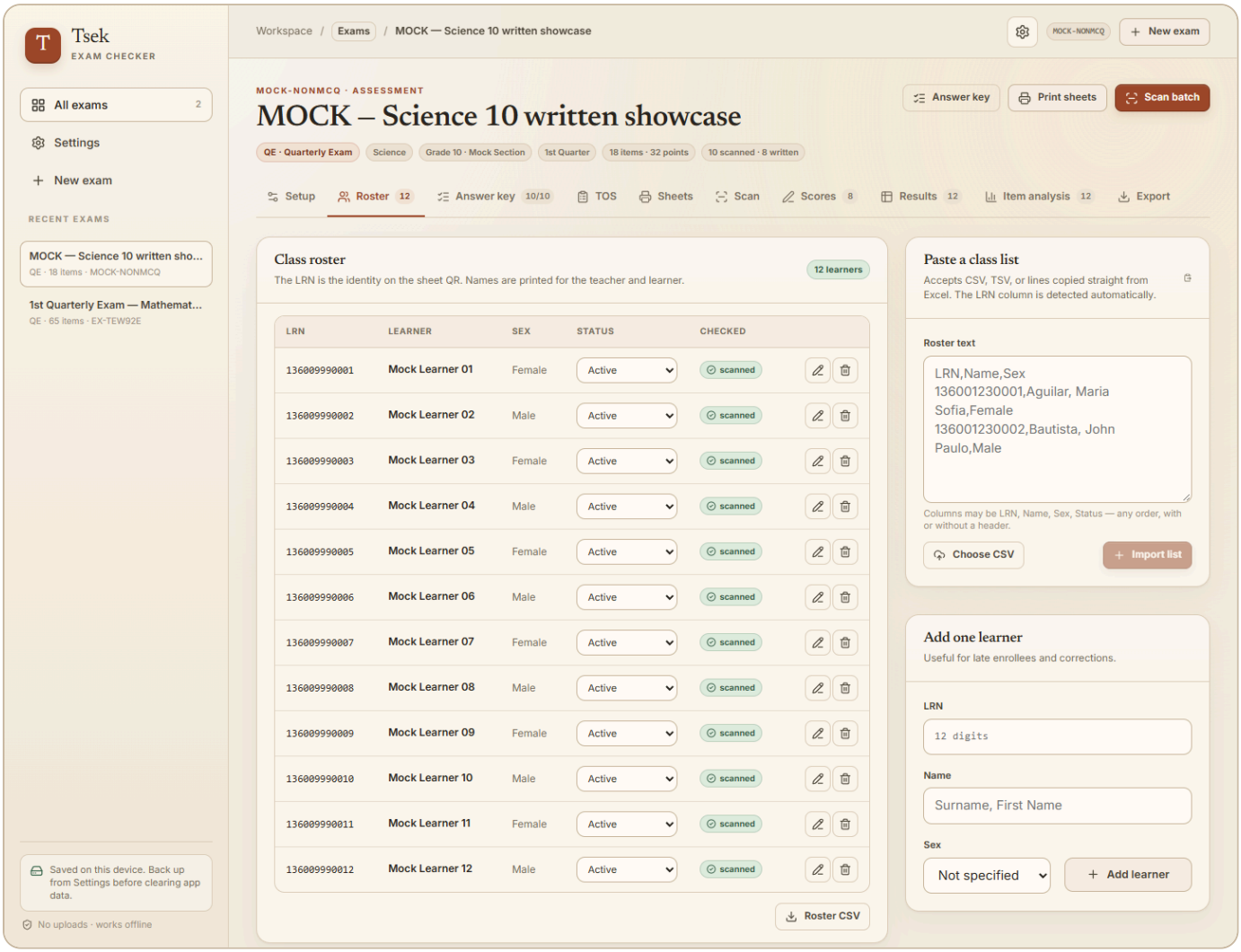


FIGURE 4 Twelve learners, each with a 12-digit LRN and a scanned badge showing which papers have been matched.

1. Open **Roster**. Paste rows straight from Excel into **Roster text**, or choose Choose CSV.
2. Columns can be **LRN, Name, Sex, Status** in any order, with or without a header. The LRN column is detected automatically.
3. Choose Import list. Existing learners with the same LRN are updated rather than duplicated.
4. For a late enrollee, use **Add one learner**: 12-digit LRN, name as *Surname, First Name*, and sex.
5. Check the **Checked** column. A green "scanned" badge means at least one paper is matched to that learner; a dash means none yet.
6. Use Roster CSV at the bottom for your own records or to hand the list to another teacher.

> **NAMES DO NOT HAVE TO MATCH EXACTLY**
>
> The QR decides who a paper belongs to. A misspelled name is cosmetic; a wrong LRN is not. If two learners share a name, the LRN keeps them apart.

5 Set the answer key

Only scanned parts need a key. Written parts are scored per learner later, so Tsek does not pretend there is one right answer.

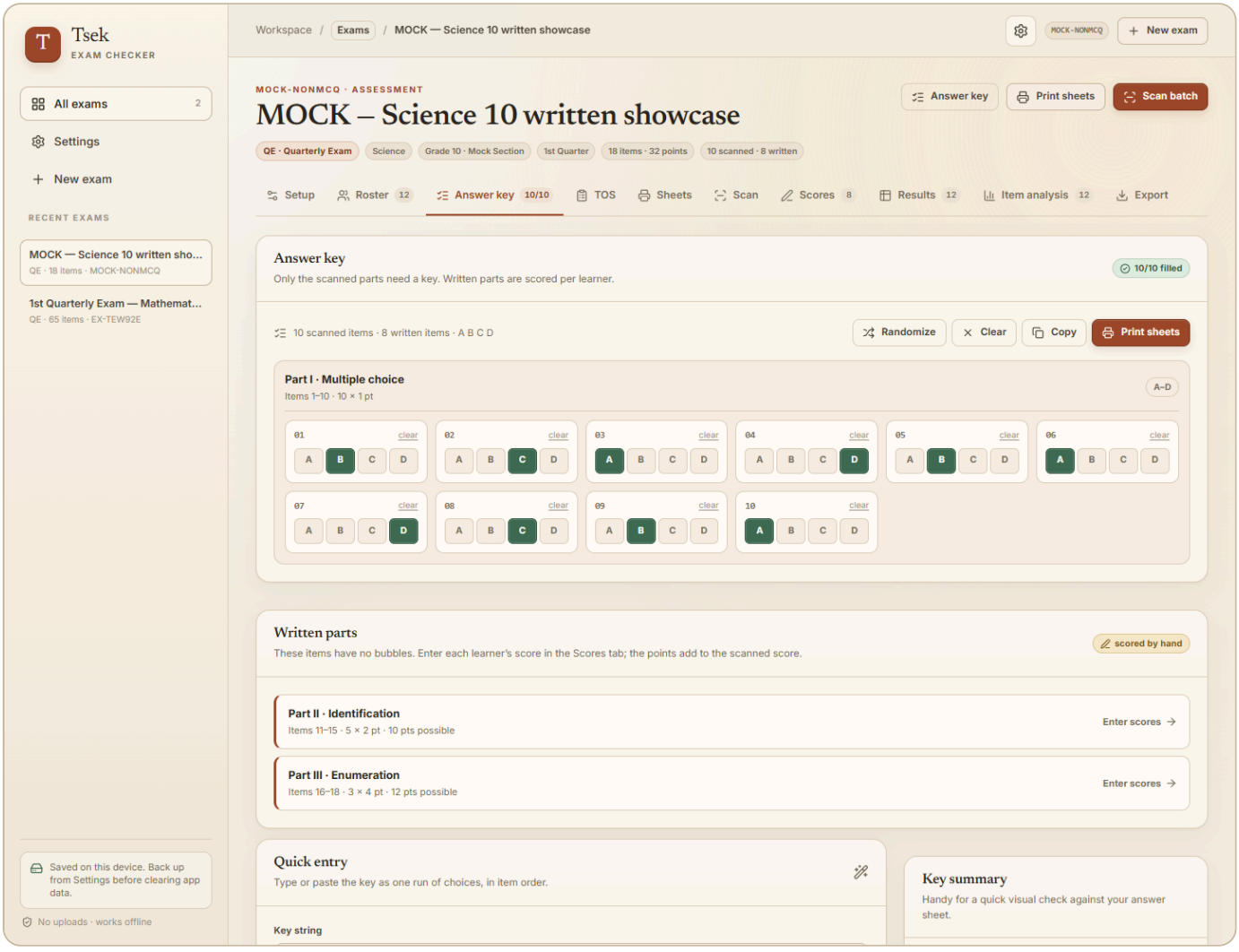


FIGURE 5 Ten keyed items in green. The badges read 10/10 filled, and the written parts list points straight to the Scores tab.

1. Open **Answer key**. Each item shows its choices; the keyed letter is dark green.
2. Click a letter to key it, or **clear** above an item to remove it. Randomize fills a valid key for practice sheets; Clear empties all scanned items.
3. For a long test, use **Quick entry** and type or paste the key as one string, in item order (for example BCDDDCADDB).

4. Use **Key summary** as a fast visual check against your answer sheet before printing.
5. Written parts appear below with their item range and points. Choose `Enter scores` to jump to the Scores tab when the papers are ready.

> **KEY CHANGES RECALCULATE SAVED SCORES**
>
> Tsek re-scores existing scans against the current key. Correct a key before reading Results or analysis; afterwards, every affected score and grade moves with it.

6 Build the Table of Specifications

The TOS is the one- or two-page table your school asks for: which competency, which Bloom level, how many items, and where they sit in the test.

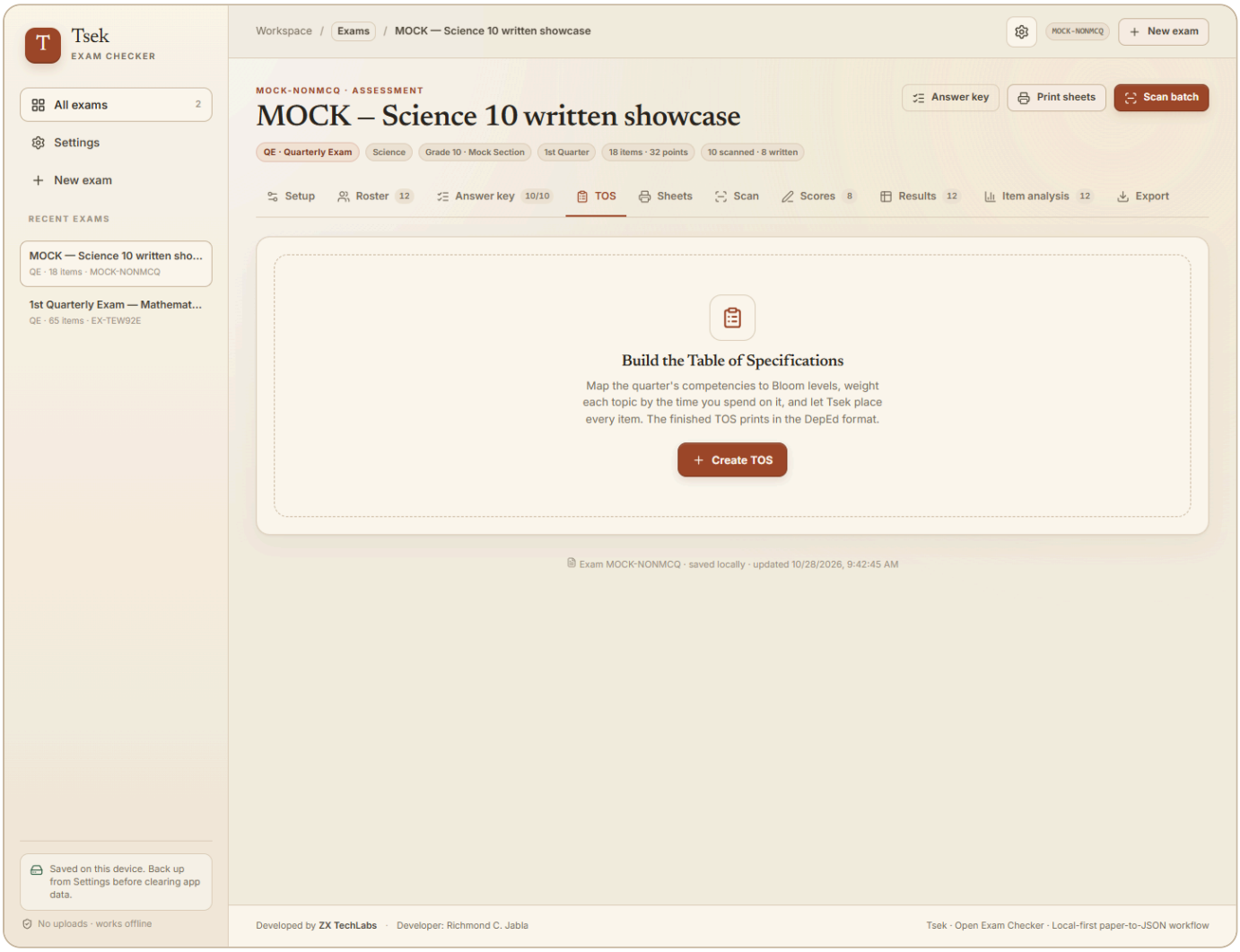


FIGURE 6 The TOS starts empty on a new exam. Create it after the structure is saved.

1. Open **TOS** and choose `Create TOS`.
2. Map each quarter competency to a Bloom level and weight it by the time you actually spend on it.

3. Tsek places the items using the part structure, so the TOS and the printed sheet cannot disagree about how many items exist.
4. Print the finished TOS in the DepEd format from the same tab.

> **ONE TOTAL, NOT TWO**
>
> The TOS total comes from Setup: scanned items plus written items. There is no separate TOS total field to keep in sync. If the total looks wrong, fix the part counts in Setup.

7 Print the answer sheets

One PDF for the class. Each sheet carries the learner's name and LRN, a QR code, and four black corner marks. The scanner uses all three.

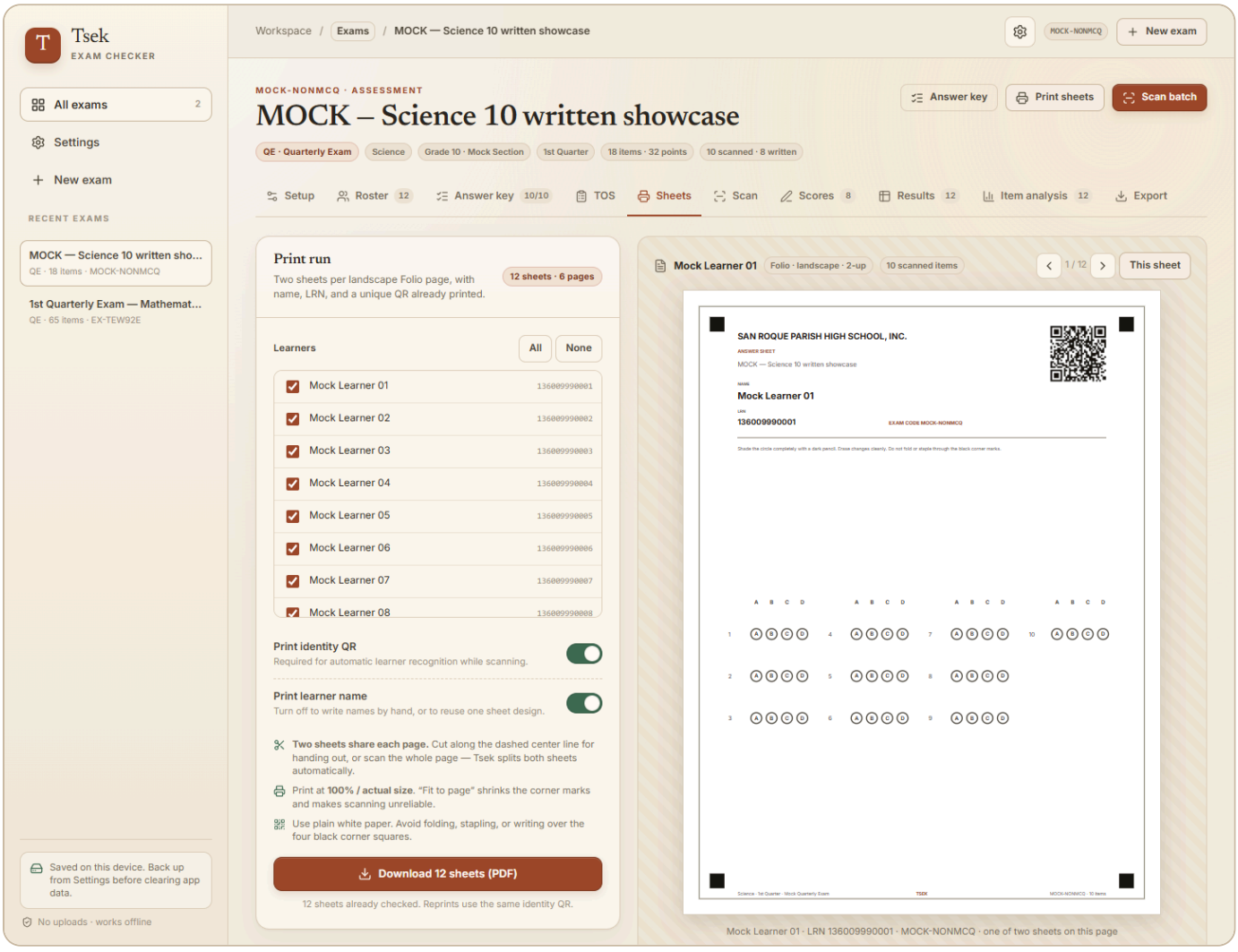


FIGURE 7 Twelve learners produce 12 sheets on 6 folio landscape pages. The preview shows one learner's half of a two-up page.

1. Open **Sheets**. Choose the learners with the checkboxes, or use `All` / `None`.

2. Leave **Print identity QR** on. It is required for automatic learner recognition. Turn off **Print learner name** only if you want identical sheets written by hand.
3. Choose `Download 12 sheets (PDF)`. The button states the exact sheet and page count.
4. In the print dialog, choose **100% / actual size**, not “Fit to page”. Use plain white paper.
5. With the default two-up layout, two portrait sheets share one landscape Folio page. Cut along the dashed centre line when handing out, or scan the whole page — Tsek splits both halves automatically.
6. Check one sheet in the preview with `This sheet` before printing the stack.

> **REPRINTS ARE SAFE**
>
> The QR belongs to the learner, not to a print run. Reprinting a lost sheet keeps the same identity, so a replacement paper scans exactly like the original. Avoid folding, stapling, or writing over the four corner squares.

8 Scan the batch

Drop the scanned PDF or the page photos. Every page is read on this device; nothing is uploaded.

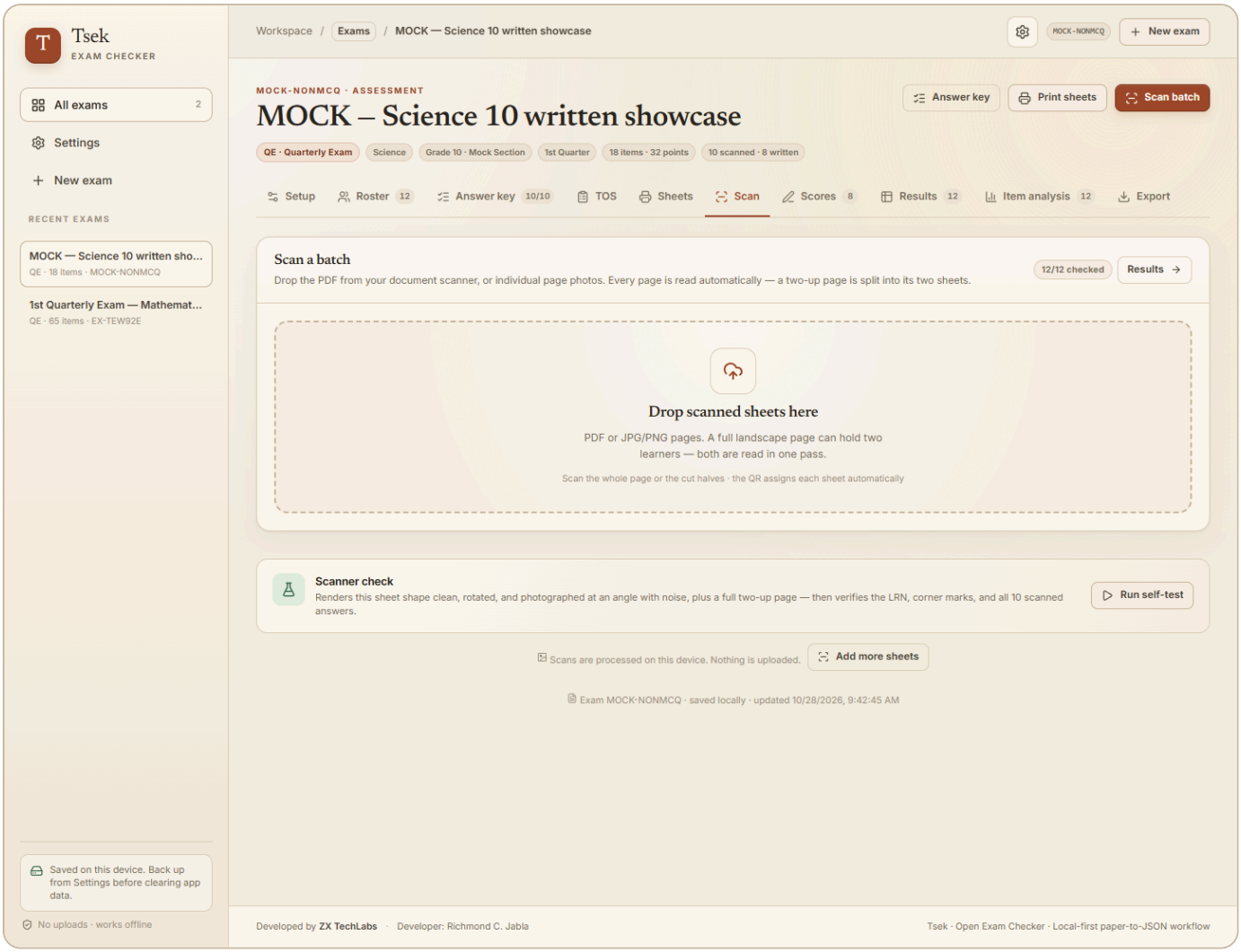


FIGURE 8 The drop zone accepts PDF, JPG, or PNG. A full landscape page holds two learners and is split in one pass.

1. Open **Scan** and drop the file into **Drop scanned sheets here**, or click to choose files.
2. A two-up landscape page is split into its two sheets. Cut halves scanned one at a time also work; the QR decides which sheet is which.
3. Watch the counter. When it reads **12/12 checked**, the whole class has been matched.
4. Open `Results` to work through anything flagged. Faint marks, double shading, and broken corners are held for review instead of being guessed.
5. If scans keep failing, run `Run self-test`. It renders the current sheet clean, rotated, and photographed at an angle, then checks the QR, corner marks, and every answer.
6. Use `Add more sheets` for late papers; new pages merge into the same class.

GOOD SCANS NEED CONTRAST, NOT PERFECTION

The scanner corrects perspective from the corner marks, so a slightly crooked page is fine. What breaks it is a cropped corner, a shadow over the QR, or “Fit to page” shrinking the whole sheet.

9 Enter the written scores

One column per written item, or one column for a pooled part. Each typed score adds directly to the scanned total.

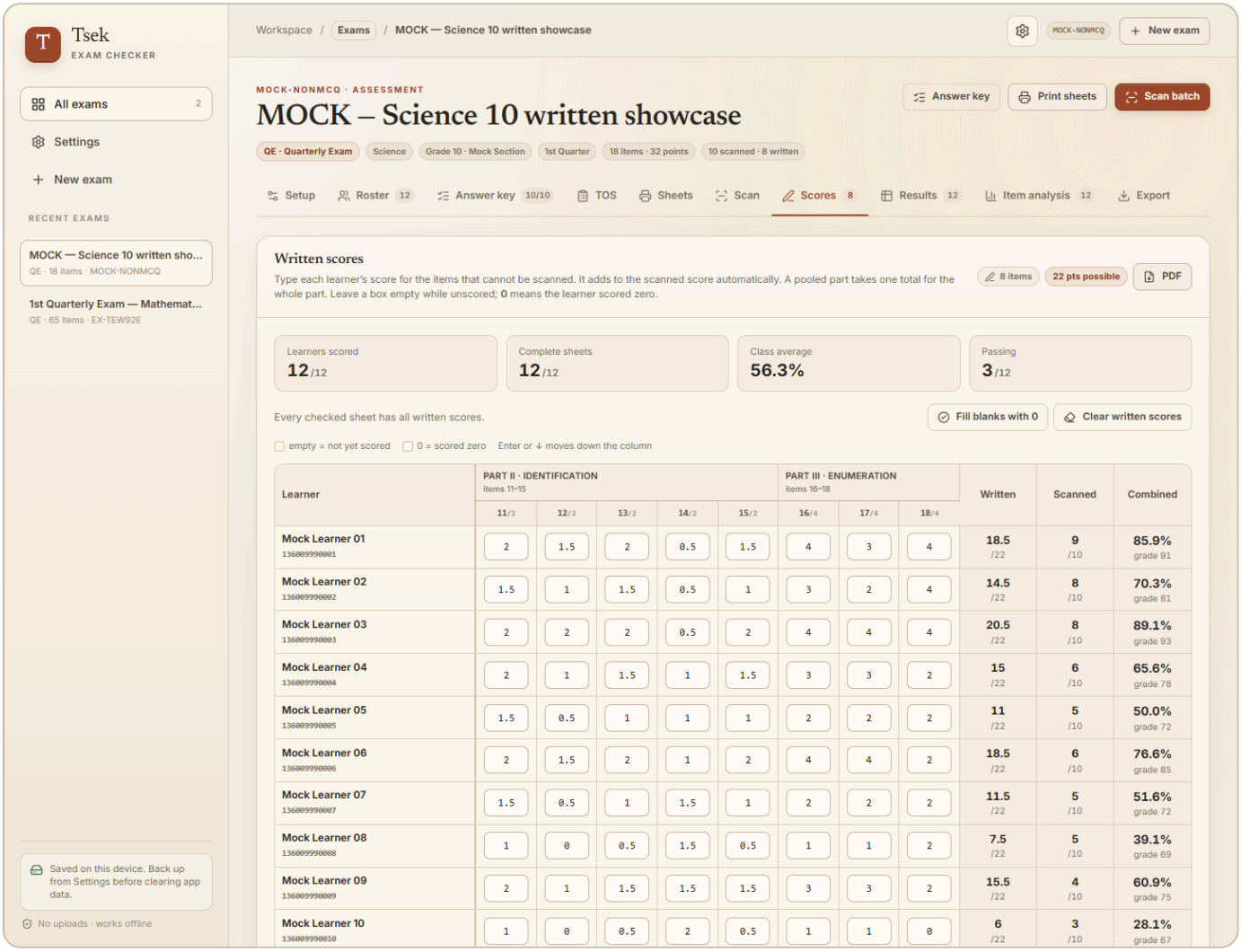


FIGURE 9 Identification is scored per item in a 2-point box; enumeration in a 4-point box. Written, Scanned, and Combined update as you type.

1. Open **Scores**. The header shows **8 items** and **22 pts possible** — the written half of the test.
2. Type a score in each box. **Enter** or the down arrow moves to the next learner in the same column.
3. Leave a box **empty** when the item is not yet scored. Type **0** when it was scored and earned nothing. The two mean different things, and the totals treat them differently.

4. When the stack is finished, use `Fill blanks with 0` only if every remaining blank really is a zero. `Clear written scores` starts the column over.
5. Check the four tiles: **Learners scored**, **Complete sheets**, **Class average**, and **Passing** at the passing percentage from Setup.
6. Read the right-hand columns: **Written** (out of 22), **Scanned** (out of 10), **Combined** as a percentage, and the transmuted **grade**.

> **HALF POINTS ARE ALLOWED**
>
> Type *1.5* or *3.5* when a rubric gives partial credit. Tsek keeps the exact value; analysis rescales it to the item's maximum points.

> **POOLED PARTS LOOK DIFFERENT**
>
> If a written part is set to **Whole part**, the item columns collapse into one wide **whole part** column. That is expected; see step 12.

10 Review the results

Results is the class list: one row per learner, with the score breakdown, grade, and review state.

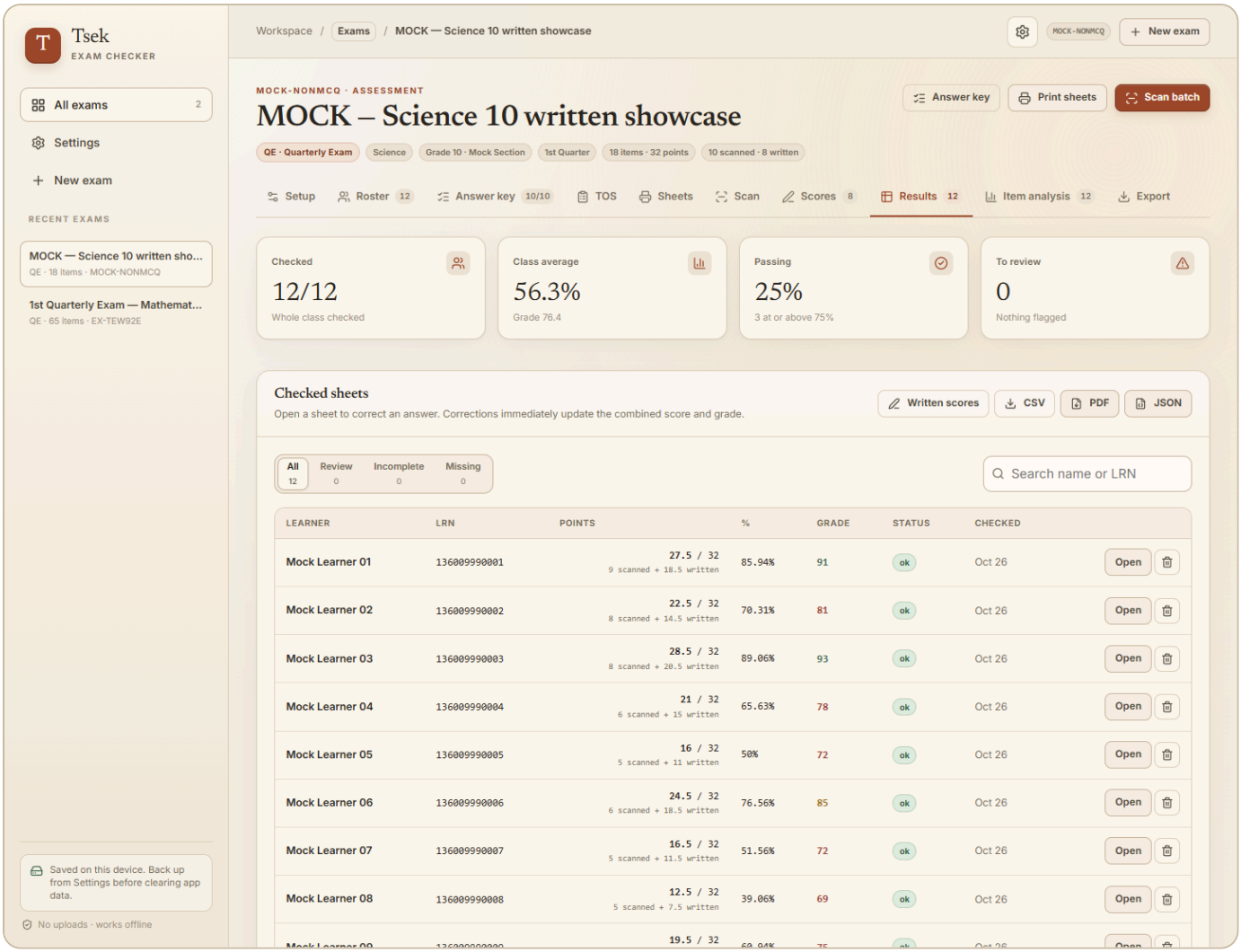


FIGURE 10 Twelve checked papers, a 56.3% class average, and 25% passing at the 75% mark. Every row is "ok" and nothing is waiting for review.

1. Check the four tiles: **Checked**, **Class average**, **Passing**, and **To review**. A complete class shows 12/12 checked and 0 to review.
2. Use the filters (**All, Review, Incomplete, Missing**) or the search box to find a learner by name or LRN.
3. Read the points cell: the top number is the total, and the small line underneath splits it, for example *9 scanned + 18.5 written*.
4. Choose Open to correct one answer. A correction immediately updates the combined score and the transmuted grade.
5. Use the header buttons for **CSV**, **PDF**, and **JSON** when you need a file rather than the screen.

WHAT "COMPLETE" MEANS

A sheet is complete when it has been scanned and every written box for that learner has a score. One empty written box keeps it incomplete, even if the scanned part is perfect.

11 Read the item analysis

Item analysis answers a different question from Results. Results says how the class did; analysis says how the test itself behaved — which items worked, which were too easy, and which did not separate strong from weak learners.

Start with the scope

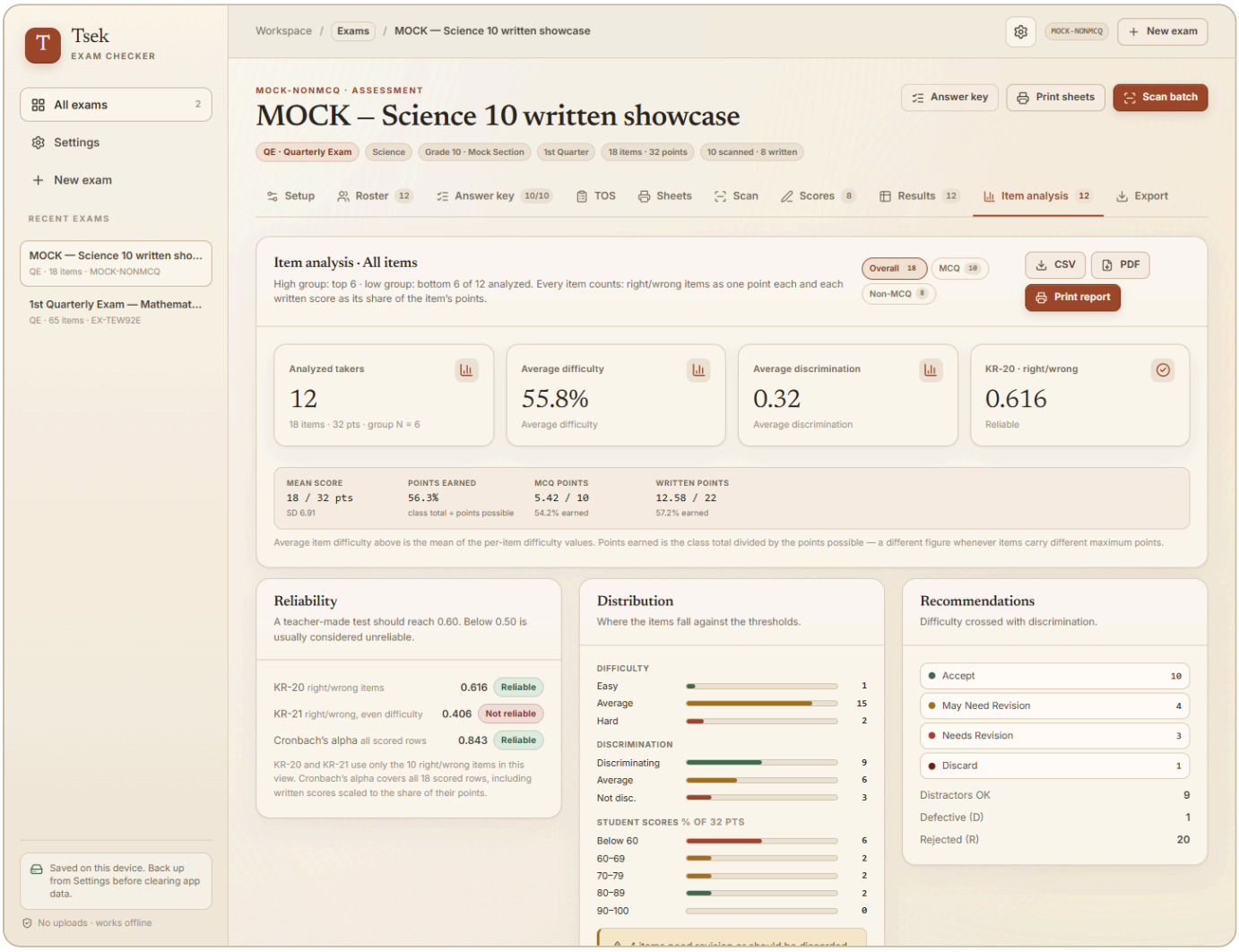


FIGURE 11 Overall analysis: all 18 items, 32 points, 12 takers. Each scope tab carries its own item count.

1. Open **Item analysis**. The scope buttons read **Overall 18**, **MCQ 10**, and **Non-MCQ 8** for this exam.
2. Use **Overall** for the whole test, **MCQ** for the scanned bubbles only, and **Non-MCQ** for the handwritten parts only.

3. Read the top cards: **Analyzed takers** (12), **Average difficulty** (55.8%), **Average discrimination** (0.32), and the reliability figure.
4. Read the score strip: **Mean score** 18 of 32, **Points earned** 5.42 of 10 scanned and 12.58 of 22 written. Points earned is the class total divided by the points possible; it is not the same as average item difficulty when items carry different points.
5. Use **CSV**, **PDF**, or **Print report** for a file that follows the same scope. An empty scope is skipped, not printed as a blank table.

The MCQ view

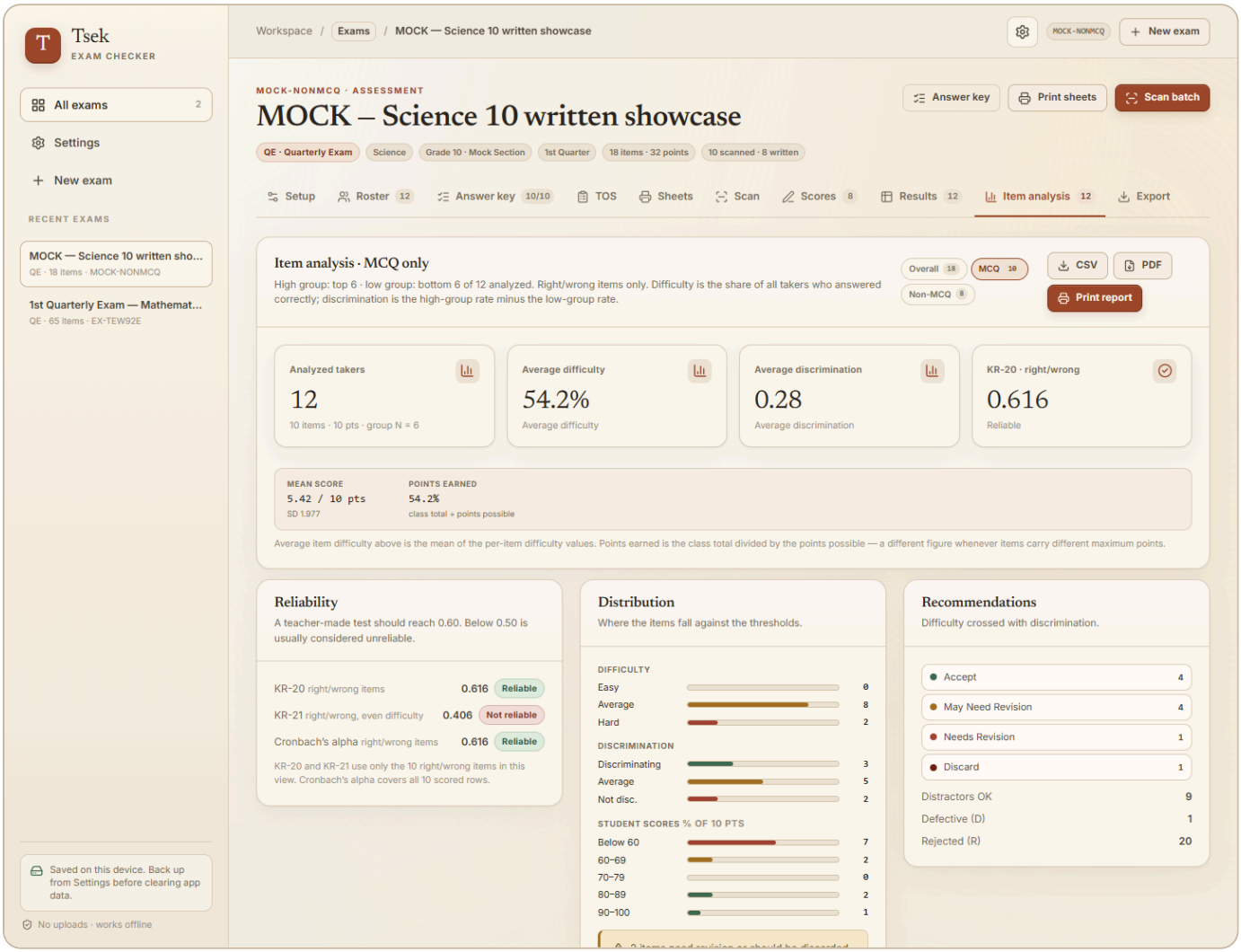


FIGURE 12 The MCQ view covers the 10 scanned items only. Written rows are excluded, so it is the fairest check of the bubble key.

Use this view when you want to judge the multiple-choice items without the handwritten parts pulling the averages around. Difficulty is the share of takers who answered correctly; discrimination is the high-group rate minus the low-group rate; reliability uses KR-20, KR-21, and Cronbach's alpha over right/wrong rows.

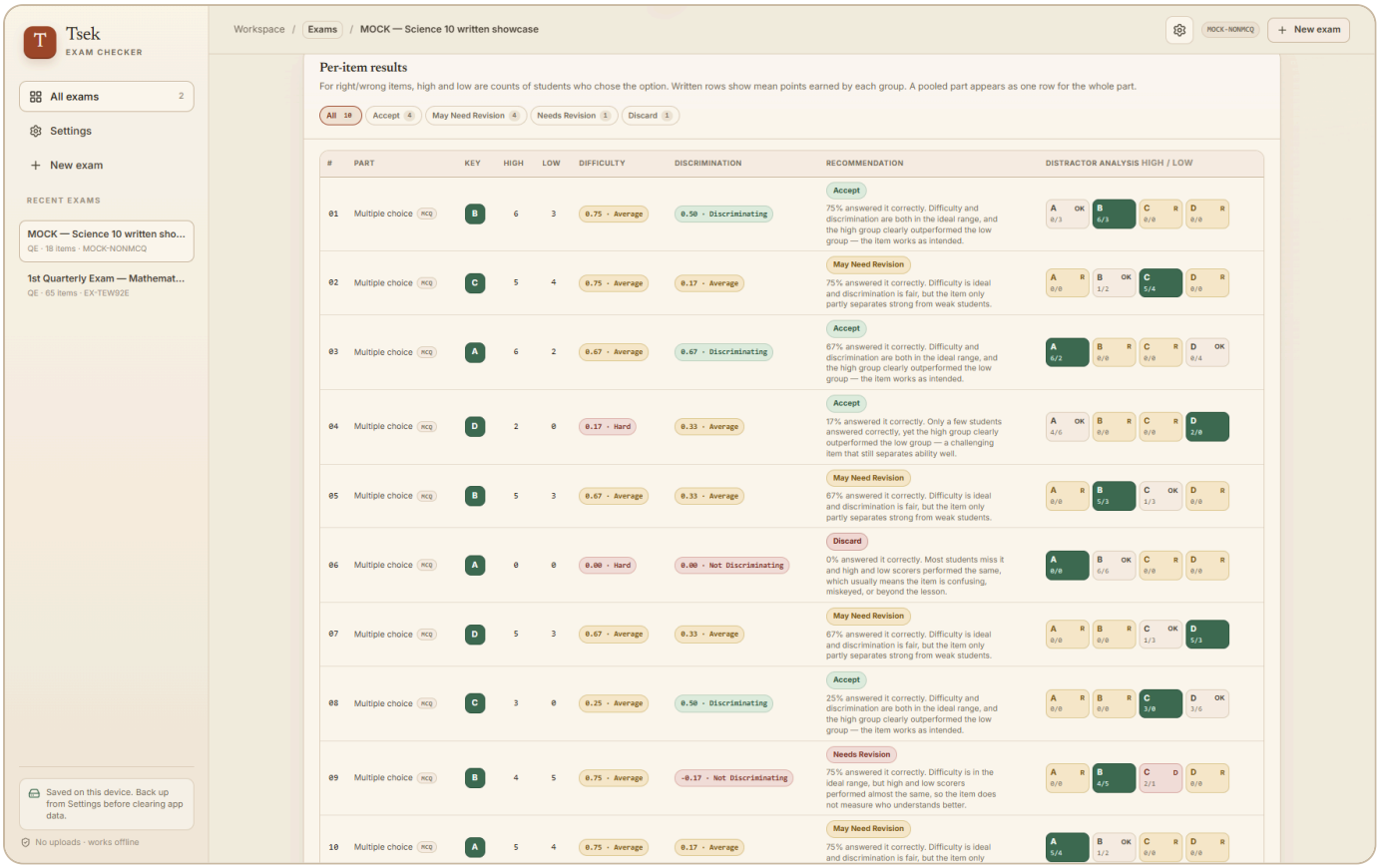


FIGURE 13 The per-item table for the MCQ view. Filter chips above match the recommendation counts; every row explains its own verdict.

The Non-MCQ view

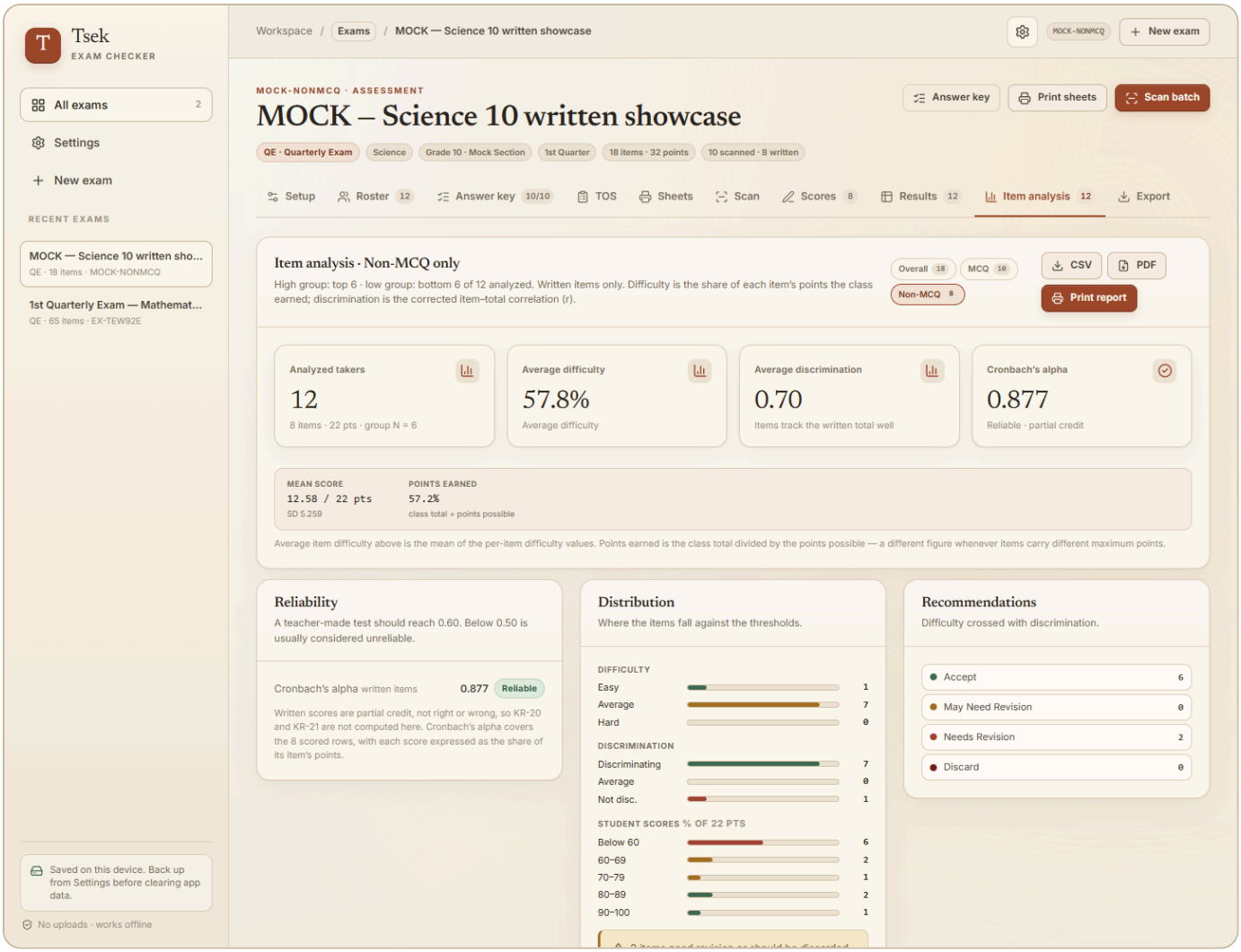


FIGURE 14 The Non-MCQ view covers the 8 written items and 22 written points. It reports Cronbach's alpha rather than KR-20.

Written items are partial credit, not right or wrong, so KR-20 and KR-21 do not apply. Difficulty here is the share of an item's points the class earned — a learner who scores 1.5 out of 2 earns 75% of that item. Discrimination is the corrected item–total correlation, which asks whether the item moves with the rest of the written score.

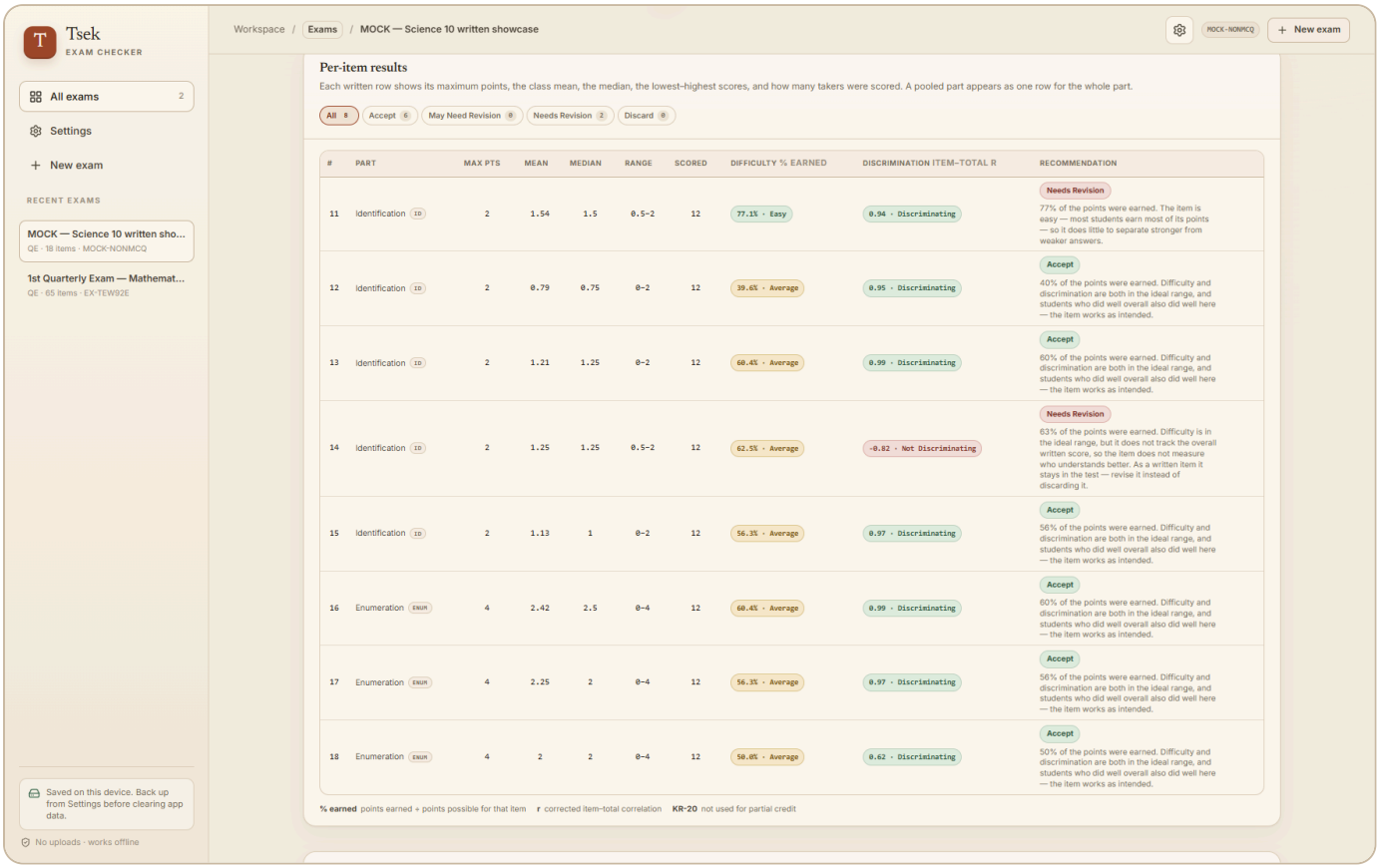


FIGURE 15 Written items 11–18. Item 14 is the clear warning: 62.5% earned but a negative correlation, so it is marked Needs Revision.

WHY SOME NUMBERS ARE DASHES

A figure that cannot be computed is deliberately left blank with the reason beside it — never filled with a zero. In this view, item 14 earned 62.5% of its points but has a negative correlation, and the recommendation says to revise it rather than discard it, because written items are not dropped the same way scanned items are.

Reliability, distribution, and verdicts

RELIABILITY

A teacher-made test should reach **0.60**. Between 0.50 and 0.60 is marginal; below 0.50 is usually considered unreliable. KR-20 and KR-21 use only right/wrong items; Cronbach's alpha also includes written rows, rescaled to each item's points.

DISTRIBUTION

Two small charts count where the items fall (Easy / Average / Hard, Discriminating / Average / Not Discriminating) and where the learners fall in 10-point score bands. They show whether the problem is a few items or the whole test.

DIFFICULTY	DISCRIMINATION	VERDICT	WHAT IT MEANS
Average (0.25–0.75)	Discriminating (above 0.40)	Accept	Works as intended; stronger learners did better.
Average	Average (0–0.40)	May Need Revision	Usable, but the item only partly tracks overall performance.
Average	Not Discriminating (0 or below)	Needs Revision	The item does not measure who understands better.
Easy (above 0.75)	Any	Needs Revision	Most learners earn most points, so it separates very little.
Hard (below 0.25)	Discriminating or Average	Accept	Hard, but it still separates strong from weak answers.
Easy or Hard	Not Discriminating	Discard	Scanned/overall views only. Written items are marked Needs Revision instead.

Thresholds are fixed across views: difficulty above 0.75 is Easy, below 0.25 is Hard; discrimination above 0.40 is Discriminating, at or below 0 is Not Discriminating. The per-item table can be filtered by verdict, but the aggregates above it always describe the whole scope.

READ THE SENTENCE, NOT JUST THE BADGE

Every recommendation is built from the same rules and then written in plain English using that item's numbers. Two items can both say “Needs Revision” for opposite reasons — one too easy, one not tracking the total.

12 Score a written part as one whole

Some handwritten answers are marked holistically: one essay score, one rubric total, one mark for the whole part. Tsek can do that without losing the items from the TOS or the analysis.

EACH ITEM

One score box per item. Best when the items are separate answers and you want per-item difficulty, discrimination, and recommendations.

WHOLE PART

One score box for the entire part, out of the sum of its item points. Best for an essay or a single holistic rubric — while every item still counts in the TOS and analysis.

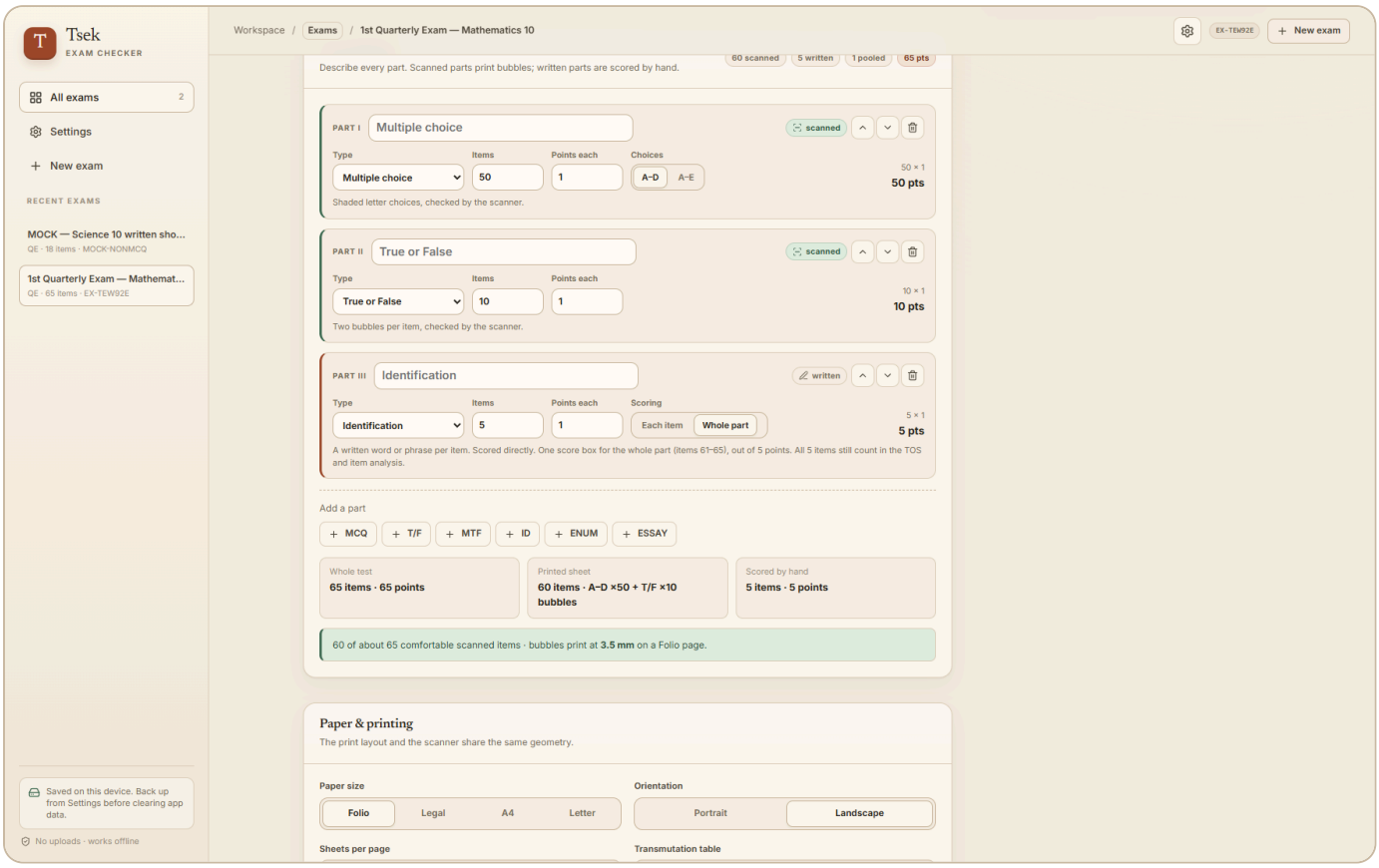


FIGURE 16 The toggle lives on the written part in Setup. The note confirms one box for items 61–65, out of 5 points, with all 5 items still counted.

1. In **Setup**, open the written part and choose **Whole part** under Scoring.
2. Read the note under the toggle. It states the item range, the single box's maximum, and that all items still count in the TOS and analysis.

3. Save the structure before entering scores, so the Scores tab rebuilds with the pooled column.

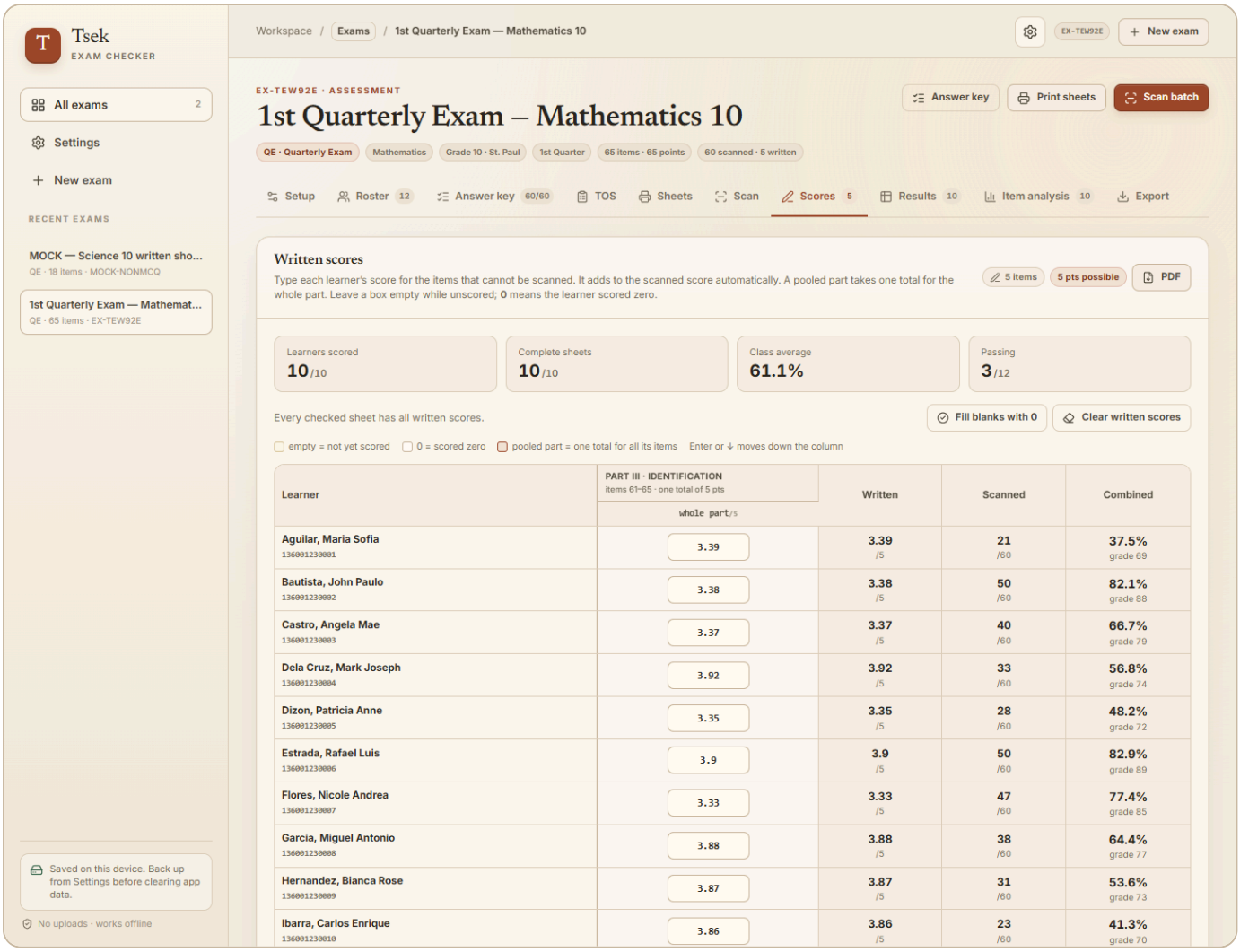


FIGURE 17 Part III now has one wide **whole part / 5** column instead of five item boxes. Decimals are accepted.

1. Open **Scores**. The pooled part shows one column headed **whole part / 5**, covering items 61–65.
2. Type the single score per learner. The written total, combined percentage, and grade update as usual.

3 The items still appear in the TOS and in item analysis; only the score entry is pooled.

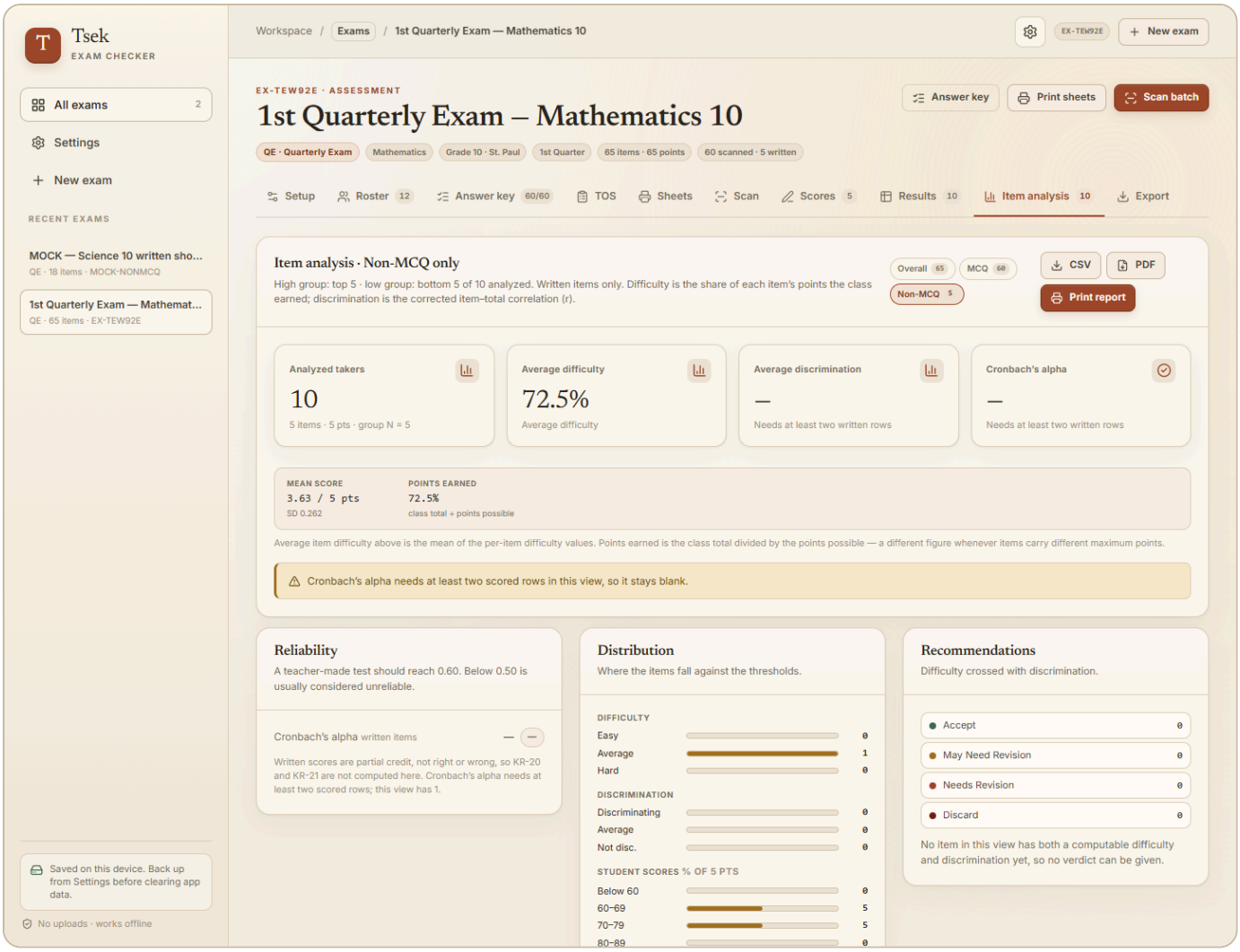


FIGURE 18 With one pooled part there is only one written row per paper, so alpha and discrimination stay blank — with the reason shown, not a fake zero.

EXPECT SOME BLANKS

A pooled part contributes one scored row per learner. Item analysis needs at least two written rows to compute alpha and discrimination, and this view has one. Difficulty (72.5%) is still meaningful because it is the part's share of its 5 points. The recommendation panel says plainly that no verdict can be given yet. That is correct behaviour, not a bug.

WHICH MODE SHOULD YOU CHOOSE?

If the learner writes five separate answers, use **Each item** — you get usable item statistics. If the learner writes one answer marked as a whole (an essay, a single rubric), use **Whole part** — the score entry matches how you actually mark it, and the TOS still counts all items.

13 Export the results and back up

Export turns the class into a file for the GradeApp, a spreadsheet, or a full archive. Every export is plain JSON or CSV; nothing leaves the device until you move the file yourself.

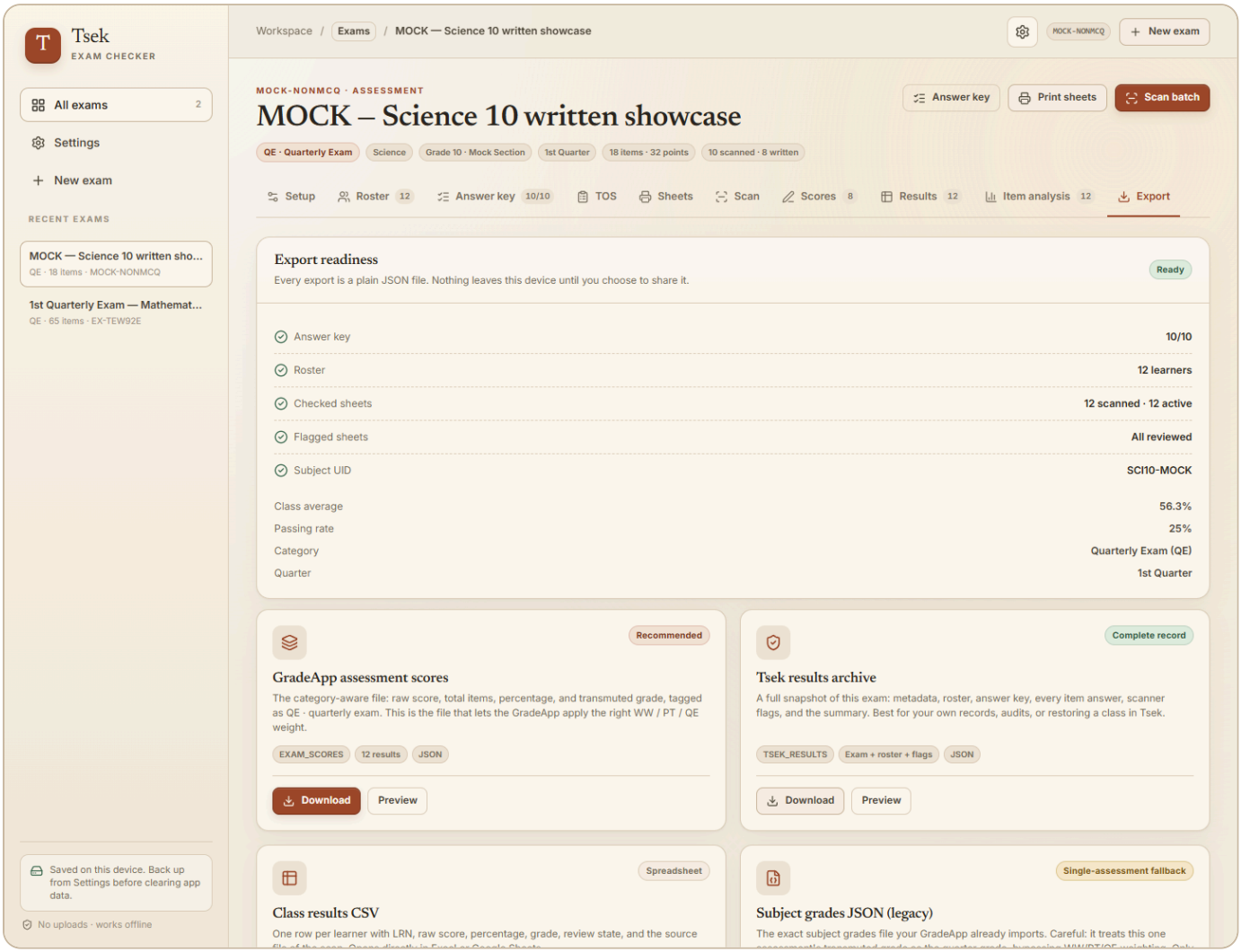


FIGURE 19 Export readiness is green when the key, roster, and scans are complete and nothing is flagged. The recommended file is the category-aware GradeApp scores.

1. Open **Export**. Confirm the readiness checklist: key complete, roster present, sheets checked, flags reviewed, Subject UID set.
2. Choose **GradeApp assessment scores**. This is the recommended file: it carries the raw score, total items, percentage, and transmuted grade tagged as WW, PT, or QE, so the GradeApp applies the right weight.
3. Choose **Tsek results archive** for a complete snapshot: metadata, roster, key, every item answer, flags, and summary. This is the file that restores a class inside Tsek.
4. Choose **Class results CSV** for a spreadsheet with one row per learner.
5. Choose **Subject grades JSON (legacy)** only when this one assessment really is the whole quarter grade. Tsek warns explicitly because the GradeApp treats this file as a complete quarter record.
6. Use `Preview` before downloading anything you are unsure about, then `Download`.

BACK UP THE WORKSPACE, NOT JUST THE EXAM

Open **Settings** → **Download workspace backup** to save all exams, learners, results, and active exam in one file. **Restore from backup** brings it back on this or another machine. Tsek keeps the last seven snapshots automatically, and every destructive action saves one first.

CLEARING APP DATA WIPES THE WORKSPACE

The browser build stores everything in local storage. Clear site data without a backup and the workspace is gone. The Windows app uses a real file in `%APPDATA%\ph.zxtechlabs.tsek\`, which survives a cache clear but not a deleted profile — keep a backup outside the machine either way.

A Labels and numbers

LABEL	WHAT IT MEANS
Analyzed takers	Complete papers used in this scope. Incomplete papers are left out.
Average difficulty	Mean of the per-item difficulty values. Higher means easier; it is not the class average score when items carry different points.
Average discrimination	How well items separate stronger from weaker learners. MCQ uses high-group minus low-group; written items use corrected item–total correlation.
KR-20 / KR-21	Reliability for right/wrong items. KR-21 assumes equal item difficulty, so it is usually lower.
Cronbach's alpha	Reliability across all scored rows, including written partial credit rescaled to each item's points.
Mean score	The class's average total in this scope, with standard deviation.
Points earned	Class total divided by points possible. With unequal item points this differs from average difficulty.
Group N	Top and bottom group size. Up to 30 takers, Tsek splits the class in half; above 30, it uses 27%.
Status ok / review	“ok” means the sheet is matched and clean; “review” means a mark was ambiguous and needs a human decision.
Incomplete	A paper is missing scanned answers or at least one written score.
— (dash)	Not computable yet. The reason is written next to it; a dash is never a zero.

B Troubleshooting

WHAT YOU SEE	WHAT TO DO
A sheet is not recognised, or its LRN is missing.	The QR is cropped, covered, or printed too small. Reprint at 100% / actual size on plain paper; keep the top-right area clear.
A paper is matched to the wrong learner.	Two sheets were mixed up, or a hand-written LRN was used with the QR off. Reprint that learner's sheet; the QR is the identity.
Items are flagged for review.	Open the sheet in Results and correct the affected answers. Faint pencil, erasures, and double shading are the usual causes; darker marks scan cleanly.
The analysis view is empty.	The scope has no items: no scanned part for MCQ, no written part for Non-MCQ. The card explains this and offers the next step. Use Overall for a test that only has one kind of item.
Reliability or discrimination shows a dash.	Not enough rows or no right/wrong items. Alpha needs two scored written rows; KR-20 needs scanned items. The note under the number states which.
No recommendation is given for an item.	Difficulty or discrimination is still missing, so no rule can be applied. The panel says so instead of guessing.
A score looks wrong after editing the key.	Tsek re-scores saved scans against the current key. Restore the correct key, or check the written-versus-scanned split in Results.
The grade looks lower than expected.	Check the transmutation table in Setup and the passing percentage. Both change the reported grade and passing count.
The printer drops faint grid lines.	Print at 100%, turn off toner-saver or draft mode, and use plain white paper. Heavy strokes are already in the sheet design; scaling is what erases them.
Windows says “Windows protected your PC”.	The build is unsigned. Choose **More info** → **Run anyway**. Code signing is not set up yet.

C Where your data lives

Tsek runs offline. Knowing where the files are is the whole backup strategy.

THING	WHERE IT IS
Windows workspace	`%APPDATA%\ph.zxtechlabs.tsek\workspace.json`
Automatic snapshots	`…\ph.zxtechlabs.tsek\snapshots\` — the last seven states
Browser workspace	Local storage key `srphs-tsek:v1` in the browser profile
Workspace backup	A JSON file saved from Settings; restores every exam, learner, and result
Sheet PDFs	Downloads folder, or wherever the Windows save dialog points
Exports	GradeApp scores, Tsek archive, class CSV, legacy subject grades

To move from the browser to the desktop app: download a workspace backup in the browser, open the desktop app, and restore that file. Scans, scores, keys, and rosters move with it.

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ZX TechLabs · Richmond C. Jabla · built first for San Roque Parish High School